

# APEX ERCOT - Day in the Life by Persona

## Purpose

This guide outlines how each core persona uses APEX through a full ERCOT day:

- Dispatch Operator
- Power Trader
- Risk Manager
- Quant Developer
- Portfolio Manager

Primary workspace screens:

- ``/workspace/dispatch``
- ``/workspace/trading``
- ``/workspace/risk``
- ``/workspace/quant``
- ``/workspace/portfolio``

Supplemental Americas screens:

- ``/americas/market``
  - ``/americas/bess``
  - ``/americas/etrm``
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## 06:00-09:00 System Start and Volatility Read

### Dispatch Operator

- Opens ``/workspace/dispatch`` with market set to ``ERCOT``.
- Validates available ERCOT assets/service types and prior stack state.
- Reviews 12-interval strip and recommendation confidence baseline.

### Power Trader

- Opens ``/workspace/trading``.
- Focuses on ``ERCOT_*`` instruments and quote responsiveness.
- Checks blotter and exposure to identify early imbalance.

### **Risk Manager**

- Opens `/workspace/risk`.
- Captures opening VaR and limit utilization.
- Sets stress baseline with current spot sensitivity.

### **Quant Developer**

- Opens `/workspace/quant`.
- Confirms ERCOT lineage and forecast coverage are present.
- Reviews strategy availability for ERCOT convergence/arbitrage themes.

### **Portfolio Manager**

- Opens `/workspace/portfolio`.
  - Reviews annualized revenue stack and contract intensity assumptions.
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## **09:00-14:00 Intraday Execution Window**

### **Dispatch Operator**

- Adjusts bid bands to reflect changing scarcity and expected spread.
- Uses recommendation engine to guide dispatch targeting.
- Validates weighted stack economics and volume caps.

### **Power Trader**

- Monitors trade flow and MTM rapidly as prices move.
- Uses quarter heatmap to avoid concentrated directional exposure.

### **Risk Manager**

- Refreshes VaR after significant order flow changes.
- Runs stress scenarios to evaluate tail expansion impact.

### **Quant Developer**

- Tracks divergence between expected and observed strip movement.
- Records anomalies tied to volatility spikes and market regime shift.

### **Portfolio Manager**

- Tests ancillary participation and PPA MW sensitivity in simulation.
- Aligns optimization recommendations with observed intraday behavior.

### **Americas Regional Screens (context augmentation)**

- `/americas/market`: compares ERCOT with other ISO pricing context.
  - `/americas/bess`: evaluates RTC+B telemetry and spread behavior.
  - `/americas/etrm`: reviews PJM/IESO context and normalization evidence.
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## **14:00-18:00 Risk and Position Governance**

### **Dispatch Operator**

- Consolidates stack updates and confirms approved action path.

### **Power Trader**

- Tightens carry exposure into close.
- Verifies execution completeness and residual risk posture.

### **Risk Manager**

- Performs structured pre-close checks:
- VaR refresh
- limit monitor review
- stress rerun at updated spot

### **Quant Developer**

- Tags model behavior windows for retraining and drift analysis.

### **Portfolio Manager**

- Reconciles current allocation against benchmark and PPA MTM impact.
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## **18:00-21:00 End-of-Day Close**

### **Dispatch Operator**

- Confirms last accepted recommendation and final stack state.

### **Power Trader**

- Finalizes blotter and carry notes for overnight desk.

### **Risk Manager**

- Publishes end-of-day risk posture and unresolved exceptions.

### **Quant Developer**

- Compiles model and data quality observations for next cycle.

### **Portfolio Manager**

- Publishes next-day optimization recommendations by component.
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## **Operating Notes for ERCOT**

- Core market data dependency: ``apex_fresh.market_ercot.lmp``.
- Core trading dependency: ``apex_fresh.trading.trades`` (market = ``ERCOT``).
- Dispatch dependencies:
  - ``apex_fresh.trading.dispatch_reference``
  - ``apex_fresh.trading.offer_bands``
- Forecast/lineage dependencies:
  - ``apex_fresh.analytics.price_forecasts``
  - ``apex_fresh.analytics.model_lineage``